Which Words Block the Machine Translation of Galen? A Reproducible Terminology Density Map

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Abstract

The obstacle to machine-translating the large untranslated Galenic corpus has recently been *measured*: a 2026 MQM study reports that passage-level terminology rarity predicts MT quality on Galen's previously untranslated pharmacological work at $r = -.97$ — the failure mode is vocabulary, not syntax. What that measurement does not supply is a queryable answer to *which* Galenic vocabulary is the blocker. We build that resource. Over the digitized Galenic corpus (First1KGreek), we run a reproducible, config-driven pipeline — extraction, odyCy lemmatization, log-likelihood keyness against a 19.68M-token non-Galen Greek reference, and LSJ cross-reference — and introduce a **four-way classification** that separates genuinely rare vocabulary from domain-specific and polysemous vocabulary, a distinction whole-corpus frequency conflates. The classification isolates, *a priori* from frequency alone, a set of common-but-technical terms (οὐσία, δύναμις, ποιότης) that rarity-based difficulty predictors cannot see by construction — a candidate failure mode orthogonal to the rarity the MT literature has measured. We release a prioritized punch list of ∼1,400 candidate terms, the full pipeline (MIT), and all data (CC-BY-4.0), each stage independently re-runnable over a different subcorpus. We claim an instrument, not a translation, and not lexicon adjudication.

1 Introduction

A large fraction of the Galenic corpus — the single biggest body of surviving ancient Greek prose — has never been translated into any modern language. The constraint is throughput, not willingness: the specialist labor required to translate technical Greek medicine does not scale, and machine translation has until recently been assumed unusable on it.

That assumption has now been quantified. A 2026 study applying MQM error annotation to machine-translated Galen reports that passage-level **terminology rarity** predicts MT quality at $r = -.97$ on the previously untranslated pharmacological work (Zainaldin et al., 2026). The practical consequence is a reframing: the task is not "improve the model's grammar" but "identify the vocabulary." If a small, well-chosen set of terms accounts for most of the degradation, then supplying those terms as constrained vocabulary is a tractable intervention where full translation is not.

No published resource says which terms those are. This paper builds the instrument panel: a reproducible map of Galen's terminological density that tells an MT researcher, or a lexicographer, *where* the hard vocabulary concentrates and *what kind* of hard each term is. We are explicit about what this is not — it is not a translation, and it is not an adjudication of what any term means. Where a term's sense is contested, we surface the term and leave the field blank for a domain expert to fill.

Contributions.

1. A reproducible, config-driven pipeline over digitized Galen, from TEI extraction through a thin-gloss punch list, with the model and thresholds pinned.
2. A **four-way classification** of candidate-hard vocabulary along two frequency axes — rarity in the medical subcorpus × rarity in the *disjoint non-medical Galen* — that distinguishes genuinely rare terms from domain-specific and polysemous ones.
3. A prioritized punch list of 1,433 candidate terms with attested example contexts, released as the scaffold for a future community-compiled Phase 3.
4. A demonstration that the frequency contrast partitions candidate-hard vocabulary, without semantic annotation, into two mechanistically distinct classes — rarity-driven difficulty (which

existing predictors capture) and common-but-sense-loaded vocabulary (which they cannot).

2 Background & Related Work

Text source. We draw on the Open Greek and Latin / First1KGreek digitization of Galen (Open Greek and Latin Project, 2016), keyed by CTS URN. The Phase 1 census (§3) fixes the work list and word counts and is released separately (Pepperman, 2026) (DOI `https://doi.org/10.5281/zenodo.21561962`).

Greek NLP. We lemmatize with **odyCy** (Kostkan et al., 2023), the Aarhus Ancient-Greek spaCy pipeline trained on Perseus and PROIEL treebanks, using the transformer model `grc_odycy_joint_trf`. odyCy is, to our knowledge, the strongest openly available lemmatizer for ancient Greek; we cite it prominently because its accuracy is the pivot on which every downstream frequency rests, and because its authors are the natural first readers of this work.

Keyness. We use log-likelihood (G^2) keyness in the standard corpus-linguistics form (Dunning, 1993; Rayson and Garside, 2000), comparing a target corpus against a reference corpus term by term.

Prior lexical work. The standard curated lexicon of Galenic medical vocabulary is Durling's *A Dictionary of Medical Terms in Galen* (Durling, 1993). It is complementary to, not in competition with, this work: Durling adjudicates what terms mean; we deliberately do not. Our map is *a priori* candidate generation from frequency structure — it surfaces which vocabulary to examine and leaves the sense to the lexicographer or the domain expert.

Relation to the motivating measurement. Zainaldin et al. (2026) is what makes this resource worth building, and it also fixes its scope. Applying MQM error annotation to machine-translated Galen, they establish that passage-level **terminology rarity** — lemmas absent from, or infrequent in, a general literary Greek reference — predicts translation quality at $r = -.97$, with critical errors concentrated in rare pharmacological terminology; conversely, they report that *common* technical terms (e.g. δύναμις, κρᾶσις) are rendered without systematic difficulty. Their analysis is passage-level — twenty passages from two works — and they explicitly invite extension to the wider corpus for the statistical power a small sample cannot supply. We take up that extension and change the unit of analysis from the passage to the term: rather than correlating a passage's rare-term ratio with its score, we map, per lemma across the digitized corpus, *where* the rare vocabulary concentrates and — via the medical-vs-non-medical contrast — *what kind* of difficulty each candidate poses. Two consequences follow. First, the contrast isolates a class their rarity predictor cannot reach by construction: common, sense-loaded vocabulary (§6). Second, because both their rare-term ratio and our counts rest on automatic lemmatization whose accuracy on Galenic medical Greek has never been reported — they lemmatize with Stanza, we with odyCy (Kostkan et al., 2023) — we measure that accuracy directly (§5) rather than assume it. The **medical-vs-non-medical comparison within a single author** is, to our knowledge, unpublished, and it is what makes the domain/polysemy distinction in §6 possible.

Corpus	Tokens	Works
Galen, medical	1,746,283	71
Galen, non-medical (disjoint)	724,353	32
Galen, whole	2,470,636	103
Non-Galen First1K reference	19,680,000 (≈)	—

Table 1: Corpus sizes used in this study.

3 Data

Census. The Phase 1 census (Pepperman, 2026) (`https://doi.org/10.5281/zenodo.21561962`) is the ground-truth work list: which editions exist, their URNs, category tags, and grc word counts. We treat it as authoritative for *which* works to process and do not re-derive it.

Subcorpora. We partition Galen into a **medical subcorpus** — works tagged pharmacology, therapeutics, pathology, pulse, semiotics, dietetics, anatomy, physiology — and a **disjoint non-medical subcorpus** (logic, protreptic, commentary, methodology, psychology, hygiene). Pseudo-Galenic attributions are excluded via an explicit config flag, not a buried default, so the choice stays auditable. Inclusion criteria are stated so a reviewer can contest any single assignment.

Sizes. The same lemmatizer model is applied to every side. This is not a detail: keyness compares

lemma inventories across corpora, so a mixed-model comparison would score model differences as if they were vocabulary differences.

4 Method

The pipeline is a linear sequence of independently re-runnable stages; each reads one input and writes one output, so any stage can be re-run without repeating the ones before it.

1. **Fetch** — download the grc TEI editions named by the census; log 404s rather than failing.
2. **Extract** — parse TEI, keep `xml:lang="grc"` edition text only, exclude apparatus (`<app>`/`<rdg>`) and restored/conjectural material. Normalize to Unicode NFC; map lunate sigma (ϲ/Ϲ → σ/ς, word-final ς); strip all-caps running headers; handle crasis/elision explicitly. A census-wordcount guardrail flags any work extracting below 70% of its recorded count.
3. **Lemmatize** — odyCy `grc_odycy_joint_trf` (spaCy 3.7.5, spacy-transformers 1.3.5, model 0.7.0, all pinned), chunked on sentence boundaries to avoid mid-sentence context loss. Per-token `(surface, lemma, pos, morph)` is retained.
4. **Frequency** — aggregate to lemma → raw count and per-10,000-token normalized count, for each corpus separately.
5. **Keyness** — log-likelihood G^2 for every medical-subcorpus lemma against the reference, with a minimum-domain-frequency floor so a lemma appearing twice cannot produce a spurious top score.
6. **LSJ cross-reference** — look up each candidate in the machine-readable LSJ (Liddell and Scott, 1940). The Perseus LSJ index is keyed in Beta Code; we build a Unicode-NFC index once at load time (Beta Code → Unicode is the less ambiguous direction and yields a human-inspectable index) and record whether an entry exists and its `<sense>` count.
7. **Punch list** — filter to thin-gloss candidates, attach example URNs and one attested context per term.

The four-way classification. The core analytic step crosses two axes: rarity in the **medical subcorpus** × rarity in the **disjoint non-medical Galen**. We use the disjoint non-medical corpus, not whole-Galen, deliberately: the medical works are 71% of the whole, so medical-frequency and whole-frequency are collinear and the contrast collapses (a pure domain term like ἀμμωνιακός, absent outside medicine, barely moves against the whole because the whole is mostly medicine). Against the disjoint non-medical corpus the contrast is real: ἀμμωνιακός reads medical 1.40 / non-medical 0.00 per-10k. The rarity threshold is a config value; we report sensitivity to it (§6). POS is restricted to NOUN/ADJ/VERB.

5 Lemmatizer Accuracy

Every frequency in this paper inherits the lemmatizer's error rate, so we report it directly rather than citing odyCy's published treebank scores — Galen's pharmacological vocabulary is out-of-domain relative to odyCy's training data, so those published numbers are an upper bound, not our number.

In the absence (to date) of a Greek-reading annotator, we bound the accuracy with three machine measurements of different epistemic character — a rule-based floor, a cross-system agreement ceiling, and a gold-standard point estimate on the rare stratum — rather than leaving it unmeasured. All three are report-only: no output of the frozen pipeline is corrected in response.

Floor: morphological admissibility. For every token we ask whether odyCy's lemma appears among the readings that Morpheus (Crane, 1991) — a rule-based analyzer sharing no training data with odyCy — admits for that surface form. On the medical subcorpus, **89.51%** of analyzable tokens carry an admissible lemma (4.41% of tokens receive no Morpheus analysis and are excluded from the denominator; they are reported, not silently pooled). Every inadmissible lemma is a near-certain error, so this is a floor on correctness; it is a slightly permissive floor, because Morpheus itself lists some variant stems as lemmas (it admits ἥπατος, a documented odyCy error for ἧπαρ, alongside the correct reading).

Agreement: independent implementations, shared training data. Against Stanza's two Ancient Greek models (Qi et al., 2020) on a 5,000-token random sample, odyCy agrees with `grc_perseus` pairwise on **82.89%** of aligned tokens, and with the two Stanza models on

94.17% of the tokens where those two agree with each other — a subset that conditioning selects for easiness. Both figures are optimistic bounds rather than accuracy: all three systems train on overlapping Universal Dependencies treebanks, so their errors are correlated, and 15.82% of the sample was lost to tokenization misalignment. We report agreement for what it is and use its disagreement set to target future human review where it is most informative.

Rare-stratum gold standard: LSJ's own Galen citations. LSJ (Liddell and Scott, 1940) cites Galen by Kühn volume and page, and each citation is a lexicographer's judgment binding an attested form at a locus to a headword. We mapped these citations onto the corpus through the editions' Kühn page breaks and accepted a token only when exactly one surface form in the cited page window is morphologically derivable from the headword per Morpheus (quoted dictionary text serves as context only, never as the scoring target, since LSJ abbreviates headwords inside entries; the full location funnel, 7,706 citations → 1,850 scored, ships with the data). On these **1,850** expert-attested loci, odyCy reproduces the lexicographer's headword for **62.32%** of tokens ignoring accentuation (Wilson 95% CI 60.09–64.50), **53.95%** under strict accent matching. LSJ-cited vocabulary is by nature the rare, lexicographically interesting stratum, so this is a stratum accuracy, not an overall one — and read against the 89.51% frequency-weighted floor, the two numbers measure, on expert-annotated data and independently of any MT system, the same rarity-driven degradation that Zainaldin et al. (2026) infer from translation error rates. The 697 mismatches decompose into: identity passthrough of an inflected form (411), garbled non-word lemmas (135), substitution of a wrong real lemma (136), and variant-headword conventions arguably not errors at all (15; scoring these as correct moves the rate only to 63.14%).

A human spot-check on the frozen random 60-token sample (seed 20260725) remains planned under the salted-control protocol shipped with the data, concentrated by the machine disagreement sets above; the machine bounds make every count below interpretable in the meantime, and we treat the rare-stratum figure as the headline caveat on the punch list (§7), whose candidates are drawn disproportionately from exactly that stratum.

Quadrant	Count	Reading
`genuinely_hard` (rare in both)	11,709	rare everyw
`polysemy_risk` (common in both)	496	common wo
`false_alarm` / domain (medical-only)	190	rare corpus-
`mostly_irrelevant` (non-medical-only)	143	not the med

Table 2: Four-way classification at threshold 1.0 per-10k ($N = 12{,}538$ lemmas).

Passthrough baseline (computed). As an independent check that Galen's vocabulary does not degrade the lemmatizer relative to ordinary Greek, we measure content-word passthrough (the rate at which a NOUN/VERB/ADJ/ADV token is emitted with `lemma == surface`, a proxy for the lemmatizer falling back to the identity). On Galen this is **26.85%**; on the 11.86M-token non-Galen reference it is **26.41%** (Δ -0.44 points). Passthrough is therefore odyCy's baseline behavior on Greek, not Galen-specific degradation — which pre-empts the natural objection that out-of-domain medical vocabulary must be breaking the lemmatizer. It is not.

Residual, systematic error classes are carried transparently into the limitations (§8) rather than silently corrected, because correcting them is a lexical judgment we have chosen not to make.

6 Results: The Four-Way Classification

Of 12,538 classified lemmas, at the threshold of 1.0 per-10k:

The off-diagonal quadrants are where the analysis earns its keep, and they survive threshold perturbation (at 0.5 / 1.5 per-10k the polysemy quadrant holds 853 / 348 members; the domain quadrant 359 / 118 — the structure is stable, not a threshold artifact).

The polysemy quadrant is what the contrast is for. Its members — δύναμις, οὐσία, ποιότης, φύσις, σῶμα, λόγος, πάθος, εἶδος, ὕλη — are high-frequency on both sides: common words that also carry Galen-technical senses. They are, by construction, invisible to any rarity signal. A term this frequent scores as *easy* under corpus-frequency rarity — which is exactly the predictor the MT study operationalizes (Zainaldin et al., 2026), and exactly what our own keyness punch list (§7) ranks. A single frequency axis cannot surface these words; the medical-vs-non-medical contrast can, because it keys on a word being *at*

home in both of Galen's registers rather than on raw scarcity. This is where the second axis earns its keep — not in re-finding the rare terms a rarity metric already flags, but in isolating the population a rarity metric is blind to.

We are deliberate about this quadrant's status — its name is `polysemy_risk`, not a verdict. That ordinary words systematically acquire specialized, register-bound senses in ancient technical medicine is well established, if in the sister Latin tradition (Langslow, 2000, ch. 3); such terms are, by their nature, the ones a reader can render in a plausible everyday sense while missing the technical one. This is a claim about the *structure of technical vocabulary*, not a finding of the MT measurement we build on — that measurement in fact reports the opposite for its own error data, finding failure concentrated in *rare* terminology while common technical terms translate cleanly (Zainaldin et al., 2026). Our claim is therefore not that these terms are hard for MT — they may not be — but that a cheap, annotation-free frequency contrast **partitions candidate-hard vocabulary into two mechanistically distinct classes**: rarity-driven difficulty, which the existing predictor already captures, and sense-loading, which it cannot see. Which class actually degrades machine translation, and by how much, is the open question §10 proposes to test.

The domain quadrant is the control. Its members — σφυγμός, δραχμή, λίτρα, ῥίζα, ὄξος, κηρός, σμύρνα, χαλβάνη, ὀπίον, ῥητίνη, ἀμμωνιακός — are materia medica and units: rare against general Greek, but ordinary within medicine. Whole-corpus keyness would flag these as "hard"; the four-way correctly files them as domain vocabulary a glossary resolves, a different and easier problem than the polysemy set. The worked case is ἀμμωνιακός at medical 1.40 / non-medical 0.00.

The practical payoff is that the classification tells an MT researcher *which kind* of hard a term is, and the two kinds want different fixes — a term glossary for the domain quadrant, sense-disambiguation for the polysemy quadrant.

7 The Punch List

The human-facing deliverable is a punch list of **1,433** candidate terms passing the thin-gloss LSJ filter, split by `lsj_tier` into **absent** (734, no LSJ entry) and **thin** (699, $\leq$1 sense line) — reported separately because they are different-quality signals: LSJ is comprehensive, so a genuine rare medical term usually has a thin entry, whereas the absent bucket also collects lemmatizer artifacts and accent mismatches. Each record carries the lemma, both frequencies, keyness, LSJ sense count, the top example work URNs, and one attested example sentence, with the identification field left empty — the scaffold a Phase 3 fills.

As a cross-validation check, **97.1%** of punch-list terms (1,391 / 1,433) fall in the `genuinely_hard` quadrant of the independent four-way classification, confirming the punch list is not contaminated by domain vocabulary.

8 Limitations

We state these plainly; under-claiming is the credibility of a resource paper.

- **Lemmatizer error is reported, not patched.** Three systematic classes remain and are handed to the reviewer rather than silently corrected: **fragmentation** (one headword split across variant lemmas, e.g. φλεβά/φλείβ → φλέψ, which distorts ranking by splitting a term's frequency); **consistently-wrong lemmas of irregular nouns** (νεύρος for νεῦρον), which the automated QC gates cannot catch by construction; and one such case surfaced only by manual audit (δεικέω → δείκνυμι). Neural-lemmatizer hallucinations — non-words with no LSJ entry and no near-neighbor surface form anywhere in the corpus — are auto-dropped (86, e.g. γαήτσρ); this is a "not a Greek word" judgment, not a lexical one. The rare-stratum gold standard (§5) quantifies these classes at scale: of 697 mismatches on LSJ-cited loci, 411 are identity passthrough, 135 garbled non-words, 136 wrong-lemma substitutions, and 15 variant-headword conventions.
- **Census caveats are inherited:** the volume approximation is unverified, and category tags are author-assigned and contestable.
- **One partial source** is documented rather than hidden: tlg035 (grc) is a 406-token Oribasius excerpt, not an extraction failure.
- **Thin LSJ gloss $\neq$ medically undefined.** The thin-gloss filter is an approximate heuristic for "under-described," not a claim that a term lacks a definition; we state it as such.

9 Availability & Reproducibility

Census: (Pepperman, 2026), DOI https://doi.org/10.5281/zenodo.21561962. Code: MIT. Data: CC-BY-4.0. Underlying Greek texts: (Open Greek and Latin Project, 2016). The lemmatizer model and all library versions are pinned; every stage is config-driven and re-runnable over a different subcorpus list, so the classification can be reproduced, audited, or retargeted.

10 Conclusion & Future Work

We have built a measured map of where Galen's translation-blocking vocabulary concentrates, and shown that a cheap frequency signal partitions that vocabulary by the *kind* of difficulty it poses, isolating a common, sense-loaded class that rarity-based predictors miss by construction.

Phase 3 compiles, for the punch-list terms, *attributed* identifications already published in the scholarly record (e.g. André on plant names, Beck's Dioscorides concordance, Berendes, LSJ) — gathering and citing, never adjudicating.

The demonstration that would close the loop: define ∼10 polysemy-quadrant terms, supply them as constrained vocabulary to an MT system, retranslate a passage, and measure the MQM quality delta against the $r = -.97$ predictor. That turns the punch list from a spreadsheet into evidence that naming the vocabulary moves the number.

A second use: adjudicating word-sense claims with corpus evidence. The same structure that separates rarity-driven from sense-loaded difficulty also answers a question that recurs outside MT — whether a given Galenic term actually carries the special sense a reader wants to assign it. Because every count is URN-keyed and partitioned by subcorpus, the map reports, for any lemma, how often it occurs and where: ordinary across Galen's registers, or concentrated and technical. That is precisely the evidence a philologist marshals by hand against the over-reading of individual words — a common term pressed to mean something esoteric — supplied here at corpus scale with a fixed, auditable provenance. We note this as a *use*, not a result: the instrument stays neutral about what any term means, but it makes the frequency-and-distribution facts that constrain such claims cheap to retrieve.

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